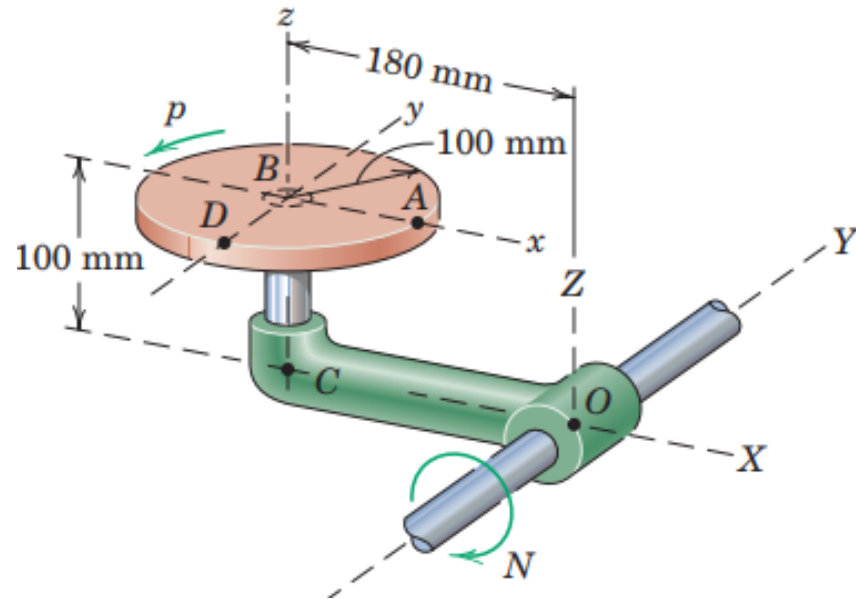


Dynamics of 3D Rigid Bodies

Dr. A. Banerjee

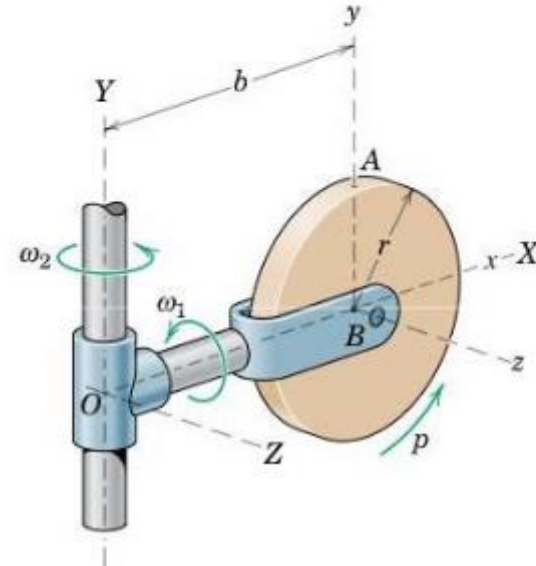
3D Kinematics

- The circular disk of 100-mm radius rotates about its z -axis at the constant speed $p = 240$ rev/min, and arm OCB rotates about the Y -axis at the constant speed $N = 30$ rev/min. Determine the velocity $\mathbf{v}$ and acceleration $\mathbf{a}$ of point A on the disk as it passes the position shown. Use reference axes x - y - z attached to the arm OCB .



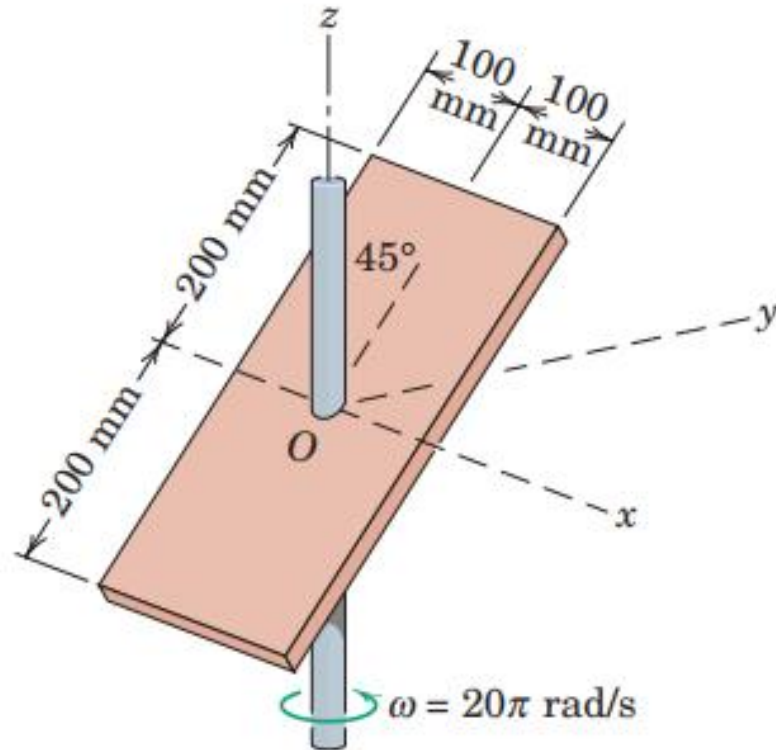
3D Kinematics

The thin circular disk of mass m and radius r is rotating about its z -axis with a constant angular velocity p , and the yoke in which it is mounted rotates about the X -axis through OB with a constant angular velocity ω_1 . Simultaneously, the entire assembly rotates about the fixed Y -axis through O with a constant angular velocity ω_2 . Determine the velocity $\mathbf{v}$ and acceleration $\mathbf{a}$ of point A on the rim of the disk as it passes the position shown where the x - y plane of the disk coincides with the X - Y plane. The x - y - z axes are attached to the yoke.



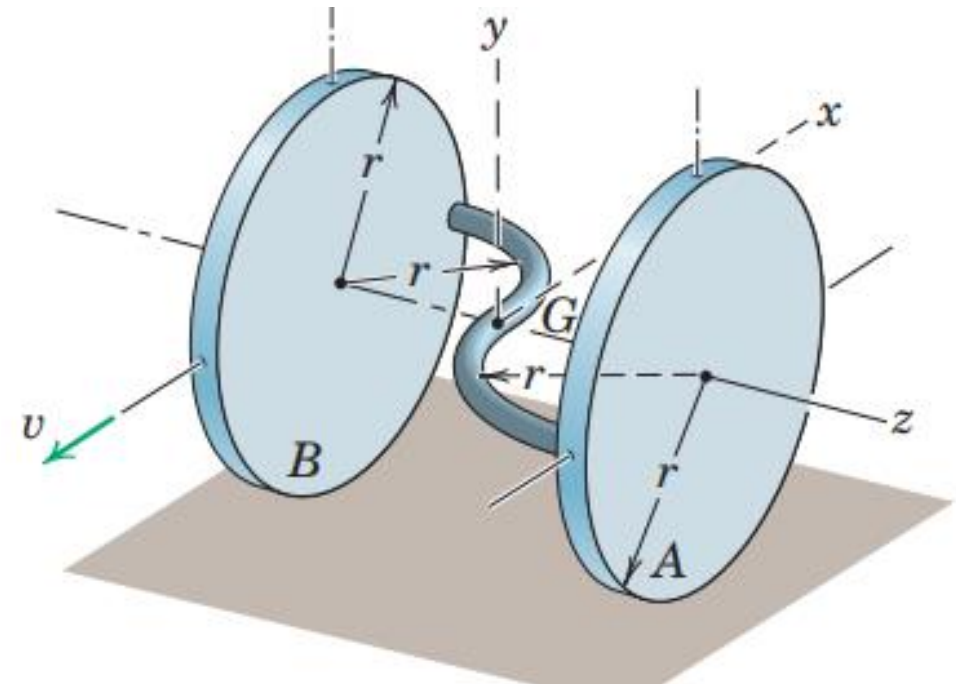
Kinetics (Example problem)

- ! The rectangular plate, with a mass of 3 kg and a uniform small thickness, is welded at the 45° angle to the vertical shaft, which rotates with the angular velocity of 20π rad/s. Determine the angular momentum $\mathbf{H}$ of the plate about O and find the kinetic energy of the plate.



Kinetics (Example problem)

The two circular disks, each of mass m_1 , are connected by the curved bar bent into quarter-circular arcs and welded to the disks. The bar has a mass m_2 . The total mass of the assembly is $m = 2m_1 + m_2$. If the disks roll without slipping on a horizontal plane with a constant velocity v of the disk centers, determine the value of the friction force under each disk at the instant represented when the plane of the curved bar is horizontal.



Kinetics

The plate has a mass of 3 kg and is welded to the fixed vertical shaft, which rotates at the constant speed of 20π rad/s. Compute the moment $\mathbf{M}$ applied to the shaft by the plate due to dynamic imbalance.

